Question ID b82a943c

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in one variable	■□□

ID: b82a943c

SAT4 M 2.1

If $7x = 28$, what is the value of $8x$?

A. 21

B. 32

C. 168

D. 224

Question ID 273b7f37

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Percentages	■□□

ID: 273b7f37

SAT4 M 2.2

Isabel grows potatoes in her garden. This year, she harvested 760 potatoes and saved 10% of them to plant next year. How many of the harvested potatoes did Isabel save to plant next year?

A. 66

B. 76

C. 84

D. 86

Question ID 3c8fdc40

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	■□□

ID: 3c8fdc40

SAT4 M 2.3

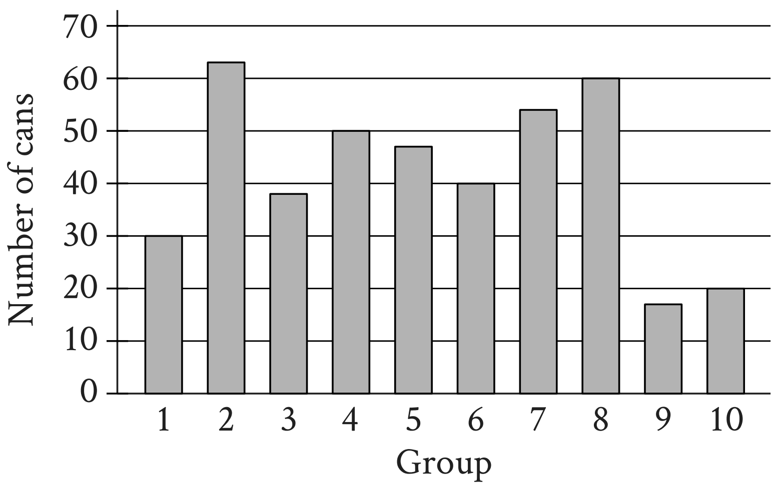
A printer produces posters at a constant rate of 42 posters per minute. At what rate, in posters per hour, does the printer produce the posters?

Question ID 820d7a73

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	■□□

ID: 820d7a73

SAT4 M 2.4



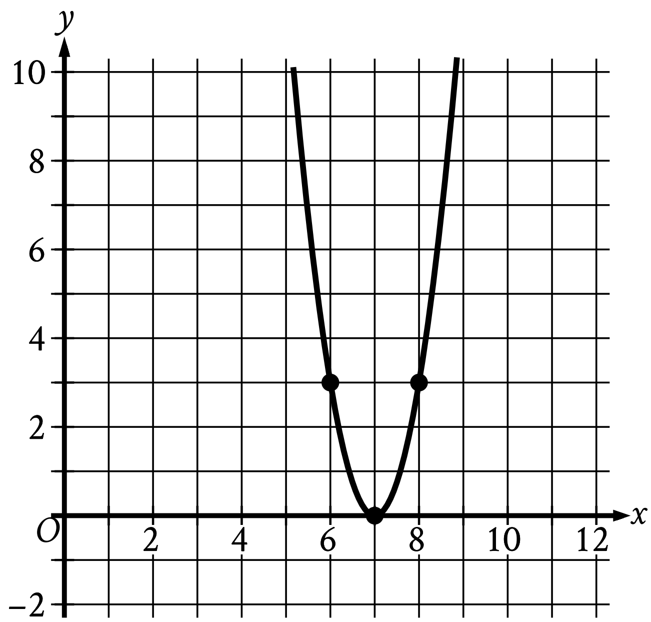
The bar graph shows the distribution of 419 cans collected by 10 different groups for a food drive. How many cans were collected by group 6?

Question ID cc2601cb

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■□□

ID: cc2601cb

SAT4 M 2.5



The x-intercept of the graph shown is $(x, 0)$. What is the value of x?

Question ID 447fa970

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear functions	■□□

ID: 447fa970

SAT4 M 2.6

The function f is defined by the equation $f(x) = 7x + 2$. What is the value of $f(x)$ when $x = 4$?

Question ID 76670c80

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	■□□

ID: 76670c80

SAT4 M 2.7

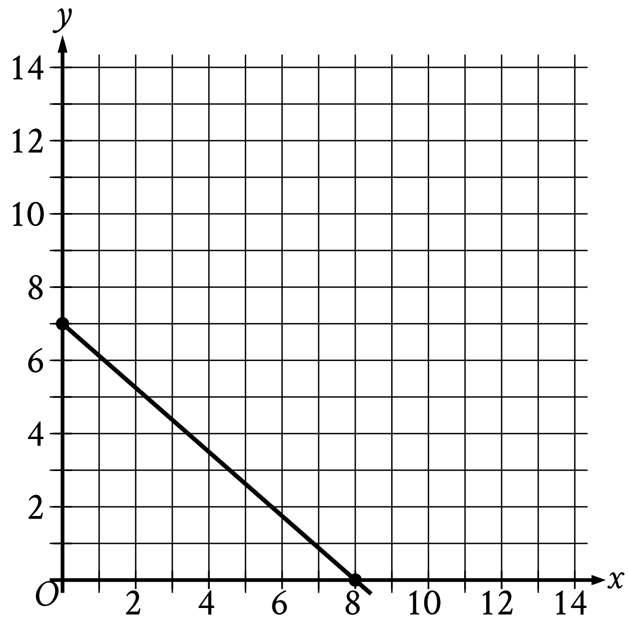
Each side of a square has a length of 45. What is the perimeter of this square?

Question ID 9d0396d4

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■■■

ID: 9d0396d4

SAT4 M 2.8



The point with coordinates $(d, 4)$ lies on the line shown. What is the value of d?

A. $\frac{7}{2}$

B. $\frac{26}{7}$

C. $\frac{24}{7}$

D. $\frac{27}{8}$

Question ID 499cb491

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Equivalent expressions	■□□

ID: 499cb491

SAT4 M 2.9

Which expression is equivalent to $5x^2 - 50xy^2$?

A. $5x(x - 10y^2)$

B. $5x(x - 50y^2)$

C. $5x^2(10xy^2)$

D. $5x^2(50xy^2)$

Question ID de362c2f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: de362c2f

SAT4 M 2.10

The function f is defined by $f(x) = 5x^2$. What is the value of $f(8)$?

A. 40

B. 50

C. 80

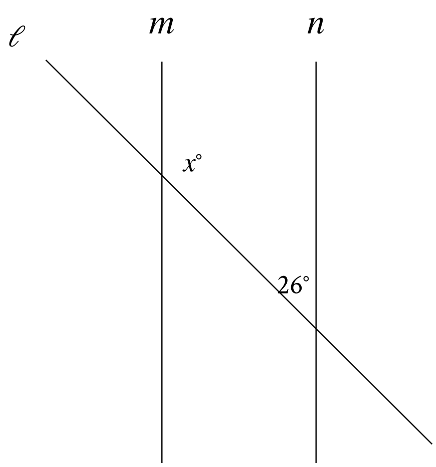
D. 320

Question ID afa3c48b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	■□□

ID: afa3c48b

SAT4 M 2.11



Note: Figure not drawn to scale.

In the figure shown, line m is parallel to line n. What is the value of x?

A. 13

B. 26

C. 52

D. 154

Question ID eccbf957

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Probability and conditional probability	■□□

ID: eccbf957

SAT4 M 2.12

Each face of a fair 14-sided die is labeled with a number from 1 through 14, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 2?

A. $\frac{1}{14}$

B. $\frac{2}{14}$

C. $\frac{12}{14}$

D. $\frac{13}{14}$

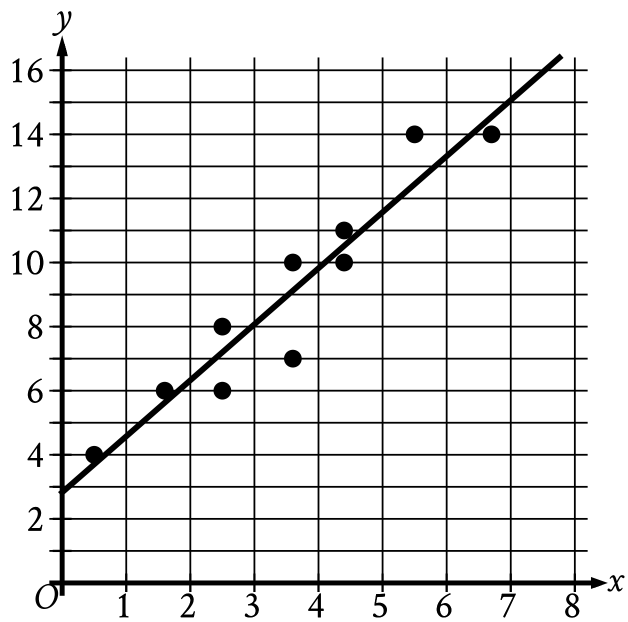
Question ID d230e963

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	■□□

ID: d230e963

SAT4 M 2.13

The scatterplot shows the relationship between two variables, x and y. A line of best fit is also shown.



Which of the following equations best represents the line of best fit shown?

A. $y = 2.8 + 1.7x$

B. $y = 2.8 - 1.7x$

C. $y = -2.8 + 1.7x$

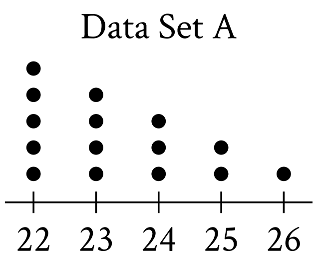
D. $y = -2.8 - 1.7x$

Question ID 4626102e

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	■■■

ID: 4626102e

SAT4 M 2.14



The dot plot represents the 15 values in data set A. Data set B is created by adding 56 to each of the values in data set A. Which of the following correctly compares the medians and the ranges of data sets A and B?

A. The median of data set B is equal to the median of data set A, and the range of data set B is equal to the range of data set A.

B. The median of data set B is equal to the median of data set A, and the range of data set B is greater than the range of data set A.

C. The median of data set B is greater than the median of data set A, and the range of data set B is equal to the range of data set A.

D. The median of data set B is greater than the median of data set A, and the range of data set B is greater than the range of data set A.

Question ID 768b2425

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■□□

ID: 768b2425

SAT4 M 2.15

Last week, an interior designer earned a total of $\$1{,}258$ from consulting for x hours and drawing up plans for y hours. The equation $68x + 85y = 1{,}258$ represents this situation. Which of the following is the best interpretation of 68 in this context?

A. The interior designer earned $\$68$ per hour consulting last week.

B. The interior designer worked 68 hours drawing up plans last week.

C. The interior designer earned $\$68$ per hour drawing up plans last week.

D. The interior designer worked 68 hours consulting last week.

Question ID c8bf5313

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■□□

ID: c8bf5313

SAT4 M 2.16

$$x = 8$$
$$y = x^2 + 8$$

The graphs of the equations in the given system of equations intersect at the point (x, y) in the *xy*-plane. What is the value of y?

A. 8

B. 24

C. 64

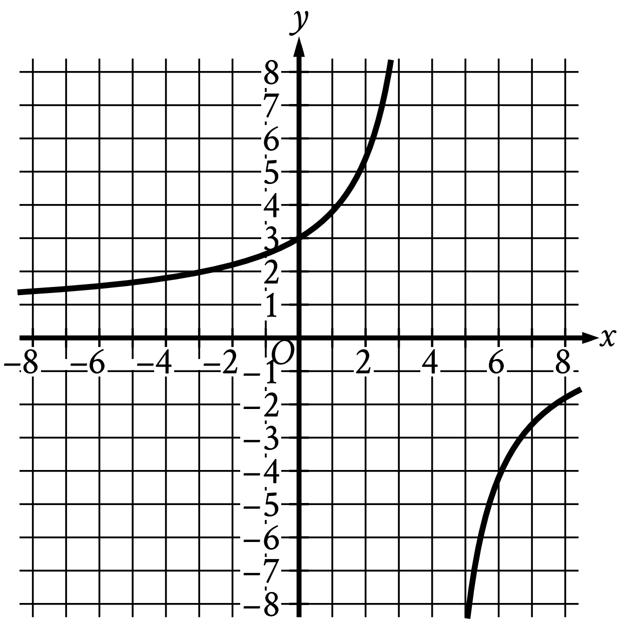
D. 72

Question ID ad376f1a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	

ID: ad376f1a

SAT4 M 2.17



The graph of $y = f(x)$ is shown in the *xy*-plane. What is the value of $f(0)$?

A. -3

B. 0

C. $\frac{3}{5}$

D. 3

Question ID 2cf7f039

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	■□□

ID: 2cf7f039

SAT4 M 2.18

The function f is defined by $f(x) = 8\sqrt{x}$. For what value of x does $f(x) = 48$?

A. 6

B. 8

C. 36

D. 64

Question ID 9f3cb472

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	■■□

ID: 9f3cb472

SAT4 M 2.19

Line t in the *xy*-plane has a slope of $-\frac{1}{3}$ and passes through the point $(9, 10)$. Which equation defines line t?

A. $y = 13x - \frac{1}{3}$

B. $y = 9x + 10$

C. $y = -\frac{x}{3} + 10$

D. $y = -\frac{x}{3} + 13$

Question ID bf4a8b6a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Systems of two linear equations in two variables	■■□

ID: bf4a8b6a

SAT4 M 2.20

A company that provides whale-watching tours takes groups of 21 people at a time. The company's revenue is 80 dollars per adult and 60 dollars per child. If the company's revenue for one group consisting of adults and children was $1,440$ dollars, how many people in the group were children?

A. 3

B. 9

C. 12

D. 18

Question ID 2cd6b22d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	■■■

ID: 2cd6b22d

SAT4 M 2.21

$$5x^2 + 10x + 16 = 0$$

How many distinct real solutions does the given equation have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Question ID e80d62c6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	■■■

ID: e80d62c6

SAT4 M 2.22

The equation $x^2 + (y - 2)^2 = 36$ represents circle A. Circle B is obtained by shifting circle A down 4 units in the *xy*-plane. Which of the following equations represents circle B?

A. $x^2 + (y + 2)^2 = 36$

B. $x^2 + (y - 6)^2 = 36$

C. $(x - 4)^2 + (y - 2)^2 = 36$

D. $(x + 4)^2 + (y - 2)^2 = 36$